

Cyber Threat Data Analysis Using Python (2016–2025)

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Tool Used: Python (Google Colab)

Abstract

Cybersecurity has become a critical concern due to the rapid growth of digital systems. This project analyzes cyber threat data from 2016 to 2025 using Python. The goal is to identify attack trends, common attack types, locations, devices, and security mechanisms using data analytics techniques.

Objectives

- Analyze cyber attack trends over 10 years
- Identify common attack types and locations
- Study devices and security mechanisms

Dataset Description

A synthetic but realistic dataset was created to ensure privacy. It includes year, attacker IP, target IP, location, type of attack, attack device, and security type.

Prompt Used

Provide a synthetic but realistic cyber threats dataset for the last 10 years (2016–2025) including attacker IP address, target IP address, location, type of attack, device of attack, and security type for data analytics projects.

Tools Used

Python, Pandas, Matplotlib, Seaborn, Google Colab

Methodology

Data loading, cleaning, exploratory data analysis, and visualization were performed using Python libraries.

Results

- Cyber attacks increased after 2020
- DDoS, Phishing, and Ransomware are dominant
- Cloud and mobile devices are common attack vectors

Conclusion

This project demonstrates the importance of data analytics in understanding and mitigating cyber threats. Python-based analysis provides meaningful insights for cybersecurity decision-making.